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	In order to safely explore beyond our planet, we need to develop methods, data bases, and validating tests for material flammability characterization, hazard reduction, and fire detection/suppression strategies for spacecraft and extraterrestrial habitats. This work addresses these needs by applying the fundamental knowledge gained from low stretch experiments to the development of a low stretch material flammability test method that simulates low gravity or partial gravity levels to better evaluate fire hazards in these environments. NASA's Cone Calorimeter Test 2 is required for major-use non-metallic panels and as an option for the retest of the flammability of a material which failed the Upward Flammability Test 1. The Cone Calorimeter is widely used ASTM standard test method because it evaluates the effects of preheating on flammability and it provides a wealth of data for evaluation. However, Test 2 has no pass/fail criteria at this point. The concept of the apparatus being developed uses the		

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low stretch geometry to simulate the conditions of the extraterrestrial environment through proper scaling of the sample dimensions to reduce the buoyant stretch we have on Earth. The apparatus uses controlled forced-air flow to augment the low stretch to levels which simulate Lunar or Martian gravity levels. In addition, the effect of imposed radiant heat flux on material flammability can be studied with the cone heater, which simulates the hazard of a small fire spreading to adjacent materials. After breadboard testing, the apparatus will be integrated into NASA's White Sands Test Facility's Atmosphere-Controlled Cone Calorimeter for evaluation as a new materials screening test method.

Primary variables include material tested, imposed forced stretch rate (effective gravity level), radiant flux level, and oxygen concentration. Data from the Apparatus will include ignition delay time, mass loss rate, heat release rate (O2 consumption), and product generation rates (CO, CO2, soot, THC). Ignition limits and material flammability limits will be measured for selected ideal/practical materials. If extinction limits are determined in Earth-based low stretch experiments as a function of the percent of heat flux conducted into the solid, we can identify critical percent heat flux values for spacecraft.

In addition to the development of the apparatus, a predictive model will be developed to predict flammability limits for a given material, using specific material properties, desired atmospheric composition, and extraterrestrial gravity environments as inputs to the prediction. The experimental and predicted results will be compared to evaluate the test method and predictive capability. We are working to reduce the fire danger to astronauts by developing an earth-bound method of evaluating materials flammability in various gravitational environments from normal gravity to microgravity, including the effects of partial gravity low stretch rates such as those found on the Moon (1/6g) or Mars (1/3g).

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

ELSA (Equivalent Low Stretch Apparatus) is proposed as a new 1g test materials screening method that is has been designed to allow us to evaluate material ignitability and flammability in spacecraft and extraterrestrial environments based upon scaling of the velocity gradient in different geometries.
This test measures a number of parameters, many of which can be used to rank materials under given conditions.

- ignition delay time, - oxygen consumption, which is used to derive heat release rates, - mass loss rates (when used with O2 consumption, we get global equivalence ratios), - CO and CO2 production rates, - critical heat flux for ignition, - soot and total hydrocarbon (not measured to date, but could be added to exhaust system using COTS hardware),

This project addresses the following organizing question(s):

3) Is there a normal gravity test that can quantify material flammability in reduced gravity either by itself or in conjunction with NASA STD 6001, Test 1?

4) How does the flammability and ignitability of materials change in high O2 concentration low pressure environments (in reduced gravity)?

To date, the fundamental similarity scaling of the stretch rate (velocity gradient) of ELSA has been demonstrated in a prototype system at NASA Glenn. An engineering unit is currently being tested at WSTF, which includes many of the exhaust diagnostics not available at the Glenn facility.

A peer-reviewed article on this test method was presented at the 30th International Symposium on Combustion [4] and is available from the PI upon request.

Preliminary Scaling Comparisons with ELSA of Independent Low Gravity Tests.

Scaling Arguments: The ELSA concept is based upon scaling analysis that demonstrates buoyant stretch and forced stretch can be equated. This means that through scaling we can assess how materials will burn over a wide range of gravity levels.

The stretch rate is simply the velocity gradient as the flow decelerates toward the fuel surface. The velocity gradient for flame spread problems has been shown to be the important flow parameter [9], and the flame is spreading deep within the boundary layer where the velocity gradient is almost constant. For forced flow velocities of up to 20 cm/s (spacecraft ventilation), the velocity gradient (stretch rate) is of the same order for small fires. Thus a test method which varies the velocity gradient (stretch rate) will capture the essential flammability behavior of flame spread as well as stagnation point fires.

FIST Data

To demonstrate this scaling with experimental results, the ignition delay results of FIST[1] at different flow rates, including natural convection, are added to the ignition delay –stretch plot by estimating the Blasius boundary layer velocity gradient at the surface for the free stream flow based on their experiment geometry. The correlation holds up very well over a wide range of velocity gradient in cm/s per cm (flame stretch rate, s-1).

While the normal cone flow field is not well characterized, the flame is stabilized near a surface with a cross flow (entrained air), and velocity gradients in a laminar boundary layer near the surface are constant. For a 5 cm surface (1/2 sample) with an average velocity gradient near the surface of 33 s-1, the average entrained cross flow would be ~20 cm/s, which is reasonable.

Skorost Data

A flammability study of different materials, including PMMA, was conducted aboard Mir in their Skorost facility [2]. Rods of PMMA were burned in a concurrent airflow parallel to the rod axis. Interestingly, the rods developed large molten balls at their burning tips, and in the low flow environment provided by Skorost, the flames were stabilized at the stagnation point of the ball. They measured rod regression rates, molten ball diameters, and flame diameters. Using that information, we can estimate the stretch rates and burning rates for these PMMA ‘lollipops’.

Regression rates of molten spheres were calculated by converting the mass loss rate (g/s) determined from the rod regression rate into the same mass loss rate from the molten sphere. Equivalent stretch rates were determined by taking

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into account both the forced stretch around the sphere (af,sphere =3/2 U/Rsphere) and the curvature-induced stretch for the small samples (acurv =3 U/R *(Rflame-Rsphere)/Rsphere). The equivalent stretch is then aequiv = af,sphere+ acurv. As the flame standoff distance (Rflame-Rsphere) becomes small relative to the radius of curvature, curvature effects become negligible. However, for the small spheres in these experiments (<1cm), the curvature effects were significant and needed to be accounted for. The comparison between Skorost results and low stretch PMMA results [3] is very good, especially considering the Skorost tests were conducted in slightly higher oxygen concentrations, which explains why the data is biased above the curve.

References

1) Roslon, Olenick, Walther, Torero, Ross, and Fernandez-Pello, “Microgravity Ignition Delay of Solid Fuels in Low Velocity Flows”, AIAA J., V. 39, No. 12, 2336-2342, (2001).

2) Ivanov, A.V., Balashov, Ye.V., Andreeva, T.V., Melikhov, A.S.; “Experimental Verification of Material Flammability in Space”, NASA CR-1999-209405, 1999.

3) Olson, S.L., and T’ien, J.S.; “Buoyant Low Stretch Diffusion Flames Beneath Cylindrical PMMA Samples”, Combustion and Flame, V.121, pp. 439-452, 2000.

4) Olson, S.L., Beeson, H.D., Haas, J.P., and Baas, J.S.; “An Earth-Based Equivalent Low Stretch Apparatus for Material Flammability Assessment in Microgravity and Extraterrestrial Environments”, Proceedings of the Combustion Institute, V30, 2004, to appear.

Plans

To continue the project through completion of a new test method, we envision the following tasks need to be conducted:

1) We need to complete the comparison of WSTF-GRC data for PMMA, so that we have confidence that our understanding of the test method is robust (ignition, flow, radiant heat, enclosure effects, etc).

2) We need to evaluate materials that may slump, swell, or display other behavior that may perturb the test. Materials that drip will be automatically eliminated (failed?) because the material will ignite when it hits the hot cone, thus disrupting the test and possibly damaging the hardware. Screens or meshes need to be tested in the inverted geometry (note: screens are used in the standard cone test for vertical samples as part of the ASTM test method).

3) We expect that via similarity scaling the 1g test will provide quantitative evaluation of a material’s reduced-gravity flammability characteristics. See figures at the end of the plan for some preliminary results of analysis to date. However, to demonstrate that more fully, we need to evaluate samples in different velocity gradient /geometries to test the stretch scaling correlation of data such as oxygen consumption, ignition delay times, critical heat flux for ignition, toxic product generation, etc. The most expedient and inexpensive way to do this over the velocity gradient range of interest is aboard the parabolic aircraft. There, we can fly low gravity trajectories at high heat flux levels (~50 kW/m2, velocity gradient 10-40 s-1). In addition, we can test the partial gravity scaling by testing samples under partial gravity parabolas with a relatively low forced convective stretch (~10 s-1). These experiments will likely use PMMA as the reference material, as is the case with the standard cone. This would most likely require a new parabolic aircraft rig buildup to accommodate the cone and large samples.

4) In addition, if/when a radiant ignition-type experiment flies on ISS, we can compare the results of the flight experiment with ground-based testing of the same material.

5) While we have been operating with the WSTF Cone facility, there are significant advantages to developing a custom ELSA facility that would eliminate some of the existing constraints of the current facility, such as the confined space of the enclosure. In addition, a custom unit would allow us to address a number of the organizing questions and deliverables: - testing in candidate atmospheres (lower ambient pressure, higher O2); - testing of detectors positioned in the exhaust system; - testing of suppression system(s) integrated into the system, tied in to the detectors if desired.

6) We need to develop a robust engineering model (as opposed to a scientifically cumbersome model) that uses material properties as input and predicts flammability behavior in the end-use environment. Simple FDS models have been developed, but a detailed study has yet to be conducted. This task may be folded into the evaluation of practical fire scenarios which is a task for the entire FPDS program. Several projects are starting to work on aspects of this. Once ELSA scaling is verified, the measured material properties would serve as input. Coordination of and funding for these efforts will be done within the FPDS program.)

7) The acceptance criteria for this test method need to be developed which address the varied exploration environments. We can address this when the scaling has been demonstrated. If possible, we should include the outcomes of the fire scenario modeling and/or performance-based risk assessments

Deliverables (schedule dependent on tbd funding)

1. Comparison and assessment of WSTF and GRC data for PMMA.

2. Work with FEANICS team to define the requirements that would have to be placed on a flight experiment that could be used in support of the stretch scaling verification. (flow characterization, gas sampling, etc).

3. Demonstration of the functional relationship between stretch rate and the measured quantities (tign, etc) for materials other than PMMA. This should be done for classes of materials to show applicability of the test method.

4. Comparison of the quantities measured in the normal gravity apparatus with similar quantities in the parabolic aircraft CIR(FEANICS) apparatus to verify low-stretch similarity with reduced gravity behavior. In other words, this will determine if the boundary-layer ignition tests are suitable to confirm the scaling. We would start with PMMA and then move to other materials, depending on the outcome of (2).

5. Planning/Design of an ELSA-type insert for the CIR-chamber aircraft rig.

Bibliography Type: Description: (Last Updated: 01/12/2026)

Articles in Peer-reviewed Journals	Olson SL, Beeson HD, Haas JP, Baas JS. "An Earth-based equivalent low stretch apparatus for material flammability assessment in microgravity and extraterrestrial environments." Proceedings of the Combustion Institute, 2005 Jan;30(2):2335-43. http://dx.doi.org/10.1016/j.proci.2004.08.044 , Jan-2005
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